

🌟 WCAG 2.1 AAA COMPLIANT ASSISTIVE TELEMETRY

NetraSense — Ambient AI Assistive Vision & Telemetry Companion

Complete Technical Specification, Architectural Dossier & Component Manual

STUDENT AUTHOR

Khirabdi Tanaya Dash

LIVE DEPLOYMENT

netrasense.vercel.app

REGISTRATION NUMBER

25BCE2067

GITHUB REPOSITORY

github.com/Kettydee/netrasense



Ambient Healthcare Philosophy: Who, Why & How

Who is it for?

Visually impaired individuals, legally blind persons, and elderly people coping with macular degeneration, glaucoma, or diabetic cataracts. It also serves as an indispensable safety monitor for family caregivers, mobility therapists, and emergency clinicians.

Why was it created?

Traditional white canes only detect hazards when physical impact occurs. Cloud-only apps create dangerous 3-second delays and suffer frequent rate limit lockouts. NetraSense delivers zero-latency on-device spatial intelligence with a calming, empathetic audio experience.

How is it used?

The user positions their device camera. NetraSense continuously analyzes the walking path, providing directional acoustic sonar beeps and natural voice feedback. In emergencies, automated collision logging alerts trusted caregivers via Supabase.



Comprehensive Tools & Technologies Inventory

NetraSense combines edge computer vision, multimodal foundation models, spatial audio acoustics, and IoT telemetry into a synchronized healthcare ecosystem.

TECHNOLOGY DOMAIN	TOOL / FRAMEWORK	SPECIFICATION	EXACT PURPOSE IN NETRASENSE
Frontend Engine	React 18 & TypeScript	v18.3, Strict Typing	Declarative, accessible component hierarchy with strict interface enforcement.
Fullstack Architecture	TanStack Router & Start	v1.5 Fullstack Route System	Zero-waterfall client-side routing, route-level preloading, and accessible page transitions.
Build & Bundling	Vite	v5.4 Rollup Bundler	Sub-second Hot Module Replacement (HMR) and optimized tree-shaken production bundles.
Design System	Tailwind CSS & Shadcn/UI	WCAG 2.1 AAA Compliant	High-contrast accessible color schemes, tactile focus states, and seamless dark/light modes.
Edge Machine Learning	TensorFlow.js & COCO-SSD	WebGL/WebGPU Hardware Accelerated	80-class object detection running directly in the browser with <35ms latency and 0 cloud costs.
In-Browser Biometrics	512D ArcFace/HOG Engine	Custom Algorithm (BROWSERFACEMATCHER.TS)	Extracts 512-dimensional facial biometric vectors locally with CLAHE lighting normalization. Matches faces in <50ms.

TECHNOLOGY DOMAIN	TOOL / FRAMEWORK	SPECIFICATION	EXACT PURPOSE IN NETRASENSE
Multimodal Foundation AI	Google Gemini Multimodal API	Gemini 2.5 Flash / 2.0 Flash / 1.5 Flash	Performs deep scene narration, medicine label reading, currency denomination OCR, and facial emotion analysis.
Spatial Acoustic Sonar	Web Audio API	AudioContext Synthesizer	Generates dynamic frequency-swept acoustic pings (300Hz to 1200Hz) to provide real-time echolocation.
Speech Synthesis & Voice	Web Speech API	SpeechSynthesis & SpeechRecognition	Enables natural speech directional alerts and hands-free voice control (e.g., <i>"find bottle"</i> , <i>"who is in front"</i>).
Cloud Database & Sync	Supabase (PostgreSQL)	Realtime WebSocket Replication	Live synchronization of obstacle telemetry streams, emergency caregiver contacts, and medical profiles.
Backend Vision Server	Python YOLO11 & Flask	Ultralytics YOLO11n, OpenCV	Optional high-speed local inference server for streaming bounding boxes, spatial coordinates, and depth maps.
Hardware IoT Telemetry	Arduino & HC-SR04 Sensor	Firmware in C++, PySerial Bridge	Microsecond-accurate hardware ultrasonic rangefinder measuring distances from 2 cm to 400 cm.
Hosting & CDN	Vercel Edge Platform	Serverless Edge Network	High-availability production hosting with

TECHNOLOGY DOMAIN	TOOL / FRAMEWORK	SPECIFICATION	EXACT PURPOSE IN NETRASENSE
			automatic HTTPS and global edge distribution.



Detailed Page-by-Page Breakdown

1. Landing & Welcome Portal (/)

The gateway welcoming new navigators, caregivers, and doctors. Features the high-contrast brand mark logo, instant theme switcher (Sun/Moon), core feature pillars (Proximity Radar, Medical ID, One-Tap SOS), and direct access links into the dashboard.

2. Authentication & Access Hub (/AUTH)

Accessible sign-in and account registration powered by Supabase Auth. Also provides an instant **"Continue as Guest / Demo Mode"** button allowing evaluators and clinicians to immediately experience all features without an account.

3. Real-Time Assistive Command Center (/DASHBOARD)

The heart of NetraSense. Displays the top gamified motivation cards (*Obstacles Dodged Today*, *Safe Distance Explored*, *Active Assistance Time*), the **Blind's Eye AI Vision Lens** with spatial zone analysis (Left, Center, Right), the **Assistive Copilot Suite** (Object Seeker, Face & Mood, Indoor Nav, Scene Reader), the **Live Environment Radar** with radial distance gauge (0 to 400 cm), and the **Caregiver Quick-Glance Panel**.

4. Emergency Contacts & Caregiver Hub (/CONTACTS)

Manages primary and secondary caregiver profiles. Users can add trusted contacts, configure collision alerts, trigger instant phone calls with one tap (via `tel:` protocol), and run simulated emergency alert tests.

5. Emergency Medical ID Card (/MEDICAL-ID)

A standardized medical card for paramedics, first responders, and good Samaritans. Highlights blood group, age, level of visual impairment (Partial, Legal Blindness, Total Blindness), critical medical allergies/notes, and primary emergency phone numbers. Includes a print-optimized layout for physical wallet cards.

6. Incident & Vision Telemetry Logs (/LOGS)

A searchable, filterable historical record of every obstacle detection and collision warning. Users and caregivers can filter by threat severity (**NORMAL** , **WARNING** , **ALARMING / COLLISION**) and export the entire dataset as a standardized CSV file.

7. Device & User Profile (**/PROFILE**)

Manages patient information, impairment classification, emergency instructions, and residential address with instantaneous Supabase cloud sync.

8. Settings & Audio Preferences (**/SETTINGS**)

Comprehensive controls for tuning the speech synthesis engine (voice alerts on/off, announce normal readings, distance threshold slider), appearance mode, YOLO server endpoint URL, and external camera stream binding.

Exhaustive Button & Interactive Control Reference

Every interactive element across the NetraSense interface is built with WCAG 2.1 AAA accessibility, tactile focus outlines, and screen-reader `aria-label` tags.

BUTTON / CONTROL	LOCATION	EXACT PURPOSE & ACCESSIBILITY ACTION
Open Camera / Start	Dashboard \$rightarrow\$ BlindsEyeLens	Requests camera permission, starts TensorFlow.js COCO-SSD detection, and establishes video frame buffer for AI tools.
Stop Camera	Dashboard \$rightarrow\$ BlindsEyeLens	Stops active video tracks, releases webcam hardware, and clears canvas overlays.
YOLO Server Mode	Dashboard \$rightarrow\$ BlindsEyeLens	Switches video pipeline to the local Python YOLO11 server (<code>/VIDEO_FEED</code>) for high-accuracy bounding box streaming.
Browser Lens Mode	Dashboard \$rightarrow\$ BlindsEyeLens	Switches video pipeline to in-browser COCO-SSD client inference (0 server dependencies, 0 latency).
Mute / Unmute Audio	Dashboard \$rightarrow\$ BlindsEyeLens	Globally toggles Web Speech voice alerts. Persisted immediately in browser <code>localStorage</code> .
Voice Selector Dropdown	Dashboard \$rightarrow\$ BlindsEyeLens	Enables user to pick their preferred synthesized voice accent, gender, and speech rate.
Camera Device Dropdown	Dashboard \$rightarrow\$ BlindsEyeLens	Selects between Front Camera, Rear Camera, or External USB/Chest-Mounted Webcam.
Voice Control (Mic)	Dashboard \$rightarrow\$ Assistive Suite	Activates continuous voice recognition listening for commands like <i>"find bottle"</i> , <i>"who is in front"</i> , or <i>"describe scene"</i> .

BUTTON / CONTROL	LOCATION	EXACT PURPOSE & ACCESSIBILITY ACTION
Find Object Tab	Dashboard \$\\rightarrow\$ Assistive Suite	Opens directional object locator with acoustic audio sonar guidance.
Object Target Chips	Dashboard \$\\rightarrow\$ Object Seeker	One-tap selectors for everyday items: <i>Water Bottle, Phone, Chair, Cup, Laptop, Backpack, Book.</i>
Face & Mood Tab	Dashboard \$\\rightarrow\$ Assistive Suite	Opens dual-tier biometric face recognition and empathetic emotion analysis suite.
Scan Face & Mood	Dashboard \$\\rightarrow\$ Face & Mood	Captures video frame, computes 512D biometric vector, checks enrolled contacts, and queries Gemini for facial mood.
Enroll Face Button	Dashboard \$\\rightarrow\$ Face & Mood	Captures reference image of a loved one, generates 512D biometric vector, and stores their name for instant recognition.
Delete Enrolled Face	Dashboard \$\\rightarrow\$ Face & Mood	Removes an enrolled contact from the local and server biometric database.
Indoor Nav Tab	Dashboard \$\\rightarrow\$ Assistive Suite	Opens indoor landmark pathfinder for navigation to doors, bathrooms, kitchens, and desks.
Destination Waypoint Buttons	Dashboard \$\\rightarrow\$ Indoor Nav	Calculates directional indoor navigation vector to chosen room destination.
Next Step Button	Dashboard \$\\rightarrow\$ Indoor Nav	Confirms progress along waypoint path and speaks the next directional walking prompt.

BUTTON / CONTROL	LOCATION	EXACT PURPOSE & ACCESSIBILITY ACTION
Describe Surroundings	Dashboard \$→\$ Scene & Currency	Uses Gemini Multimodal Vision to provide an audio description of the room layout, furniture, and pathway hazards.
Read Currency / Text	Dashboard \$→\$ Scene & Currency	Inspects banknotes or medicine prescription labels and speaks out denominations and dosage instructions.
Repeat Spoken Speech	Dashboard \$→\$ Scene & Currency	Re-speaks the latest AI description without re-querying the model.
Simulate Sensor Reading	Dashboard \$→\$ Live Radar	Generates a simulated obstacle reading (0 to 400 cm) to test audio alerts, radial gauges, and database logging.
Broadcast SOS Button	AppShell Sidebar & Contacts	High-visibility emergency button alerting caregivers and redirecting to the emergency contact hub.
Call Contact Button	Dashboard & Contacts Page	Initiates direct cellular telephone call to the primary caregiver via tel: URL scheme.
Print Medical ID Card	Medical ID Page	Formats and triggers the browser print dialog to print a physical wallet card.
Export CSV Button	Incident Logs Page	Downloads complete telemetry history into an offline .csv spreadsheet for review.
Theme Toggle (Sun/Moon)	Top Header & AppShell	Switches smoothly between High-Contrast Dark Mode and Clean Light Mode.



Dual-Tier Biometric Face Recognition Architecture

NetraSense features a resilient hybrid face identification engine that balances instant offline recognition with rich empathetic understanding:

⚡ Tier 1: In-Browser 512D Biometrics

Runs locally inside `BROWSERFACEMATCHER.TS`.
Divides a normalized 128×128 face crop into an 8×8 cell grid with 8 gradient orientation bins. Normalizes lighting with CLAHE and matches via Cosine Distance in **<50ms** with **0 network latency**, **0 token cost**, and **0 rate limits**.

💖 Tier 2: Multimodal Empathy & Mood

Uses a resilient fallback chain (`GEMINI-2.5-FLASH` $\rightarrow$ `GEMINI-2.0-FLASH` $\rightarrow$ `GEMINI-1.5-FLASH`) to detect facial expressions, smiles, fatigue, or distress, speaking warm prompts such as: *"Your friend Sarah is smiling warmly."*

🛡️ Cooldown & Fallback Safety

Includes built-in 3-second pacing and an automatic 10-second cooldown recovery mechanism to prevent API exhaustion while ensuring uninterrupted local biometric recognition.



Submission Dossier & Author Certification

STUDENT CANDIDATE

Khirabdi Tanaya Dash

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LIVE DEPLOYED WEBSITE

<https://netrasense.vercel.app>

GITHUB REPOSITORY

<https://github.com/Kettydee/netrasense>

TARGET GOOGLE DRIVE FOLDER

[CLAUDE CODE Submission Folder](#)

This dossier certifies the design, architectural development, and production deployment of NetraSense. To generate a PDF copy for Google Drive submission, click the **Print / Save as PDF** button at the top of this page or press **Ctrl + P** in your browser.